

Team LONG PULL — 2026-07-08. CASEY’S METHODOLOGY: describe the BST-native Standard Model; the differences from observed are DATA to derive geometrically, NOT errors. The mass “floor” was a scheme error; $d(\nu)$ may be the native spectrum. Force the positions; CP from $N_c=3$. date + SOD.

The organizing frame (Casey)

“Describe the Standard Model based on D_{IV}^5 , then describe how any discrepancies in the BST Standard Model are derived directly from the geometry. The differences when we use BST values are NOT errors — they are DATA to be explained.”

The BST-native SM gives its own values; the difference from observed (the λ QCD dressing) is DATA to derive from BST’s own structure, not a failure and not imported λ QCD. This is the mass-sector program now.

Progress this round (noted)

- Scheme issue QUANTIFIED (Elie 4593, Grace): $\{1, 20, 900\}$ mixes scales; scale-consistent $MS\text{-bar} = \{1, 20, 1028\}$ ($m_b/m_s \approx 51$, $m_b/m_d \approx 1028$); constituent spread only $\sim 14\times$. The b-ratios carry $\sim 12\%$ λ QCD baggage. $m_s/d = 20 = \text{rank}^2 \cdot n_c$ survives clean (both at 2 GeV, scale-invariant).
- Floor RETRACTED with numbers (Grace): $895 \times (MS\text{-bar})$ vs $14 \times (\text{constituent}) = \text{factor-60}$ of pure λ QCD running. The “floor” scored BST against a λ QCD-laden target.
- $d(\nu)$ may be the NATIVE spectrum (Grace’s tell): $d(\nu)$ gave $s/d = 1.67 \approx \text{constituent} \sim 1.6$, NOT $MS\text{-bar} 20$. Graded against the wrong answer key. Partial un-retraction — re-examine.
- CP-from- $N_c=3$ (Lyra + Casey): the geometric MECHANISM for KM — 3 generations at cube-root/ 120° (complex) $\rightarrow$ CP exists (Jarlskog nonzero, generic); handedness = triangle orientation = matter/antimatter; the three = colors = generations = corners ($\mathbb{Z}_3$). Novel structural candidate.

PRIMARY — force the generation positions $\rightarrow$ native spectrum + mixing · LYRA + GRACE

Force the positions ($r_2=1/\text{rank}^2$ OR the cube-root $\mathbb{Z}_3$ structure) $\rightarrow$ the native mass spectrum (spacing) + the mixing (overlaps), together (one gate). - Compare the native spectrum to the CONSTITUENT scheme (Grace’s tell), NOT $MS\text{-bar}$. If $d(\nu)$ /the positions give the constituent-like native spectrum, the masses are *close* — and the $MS\text{-bar}$ difference is the dressing (DATA). - Higher-leverage than α root-lock #2 (Grace’s own read) — closes masses + mixing at once.

THE MASS PROGRAM (Casey’s methodology, three steps) · GRACE + ELIE

1. BST-native spectrum — the positions/ $d(\nu)$ give it; compare to CONSTITUENT ($\sim 14\times$ spread), not $MS\text{-bar}$.
2. BST-native dressing — the difference native $\rightarrow$ observed = the λ QCD-analog, derived from BST’s OWN structure (confinement = Hamming(7,4,3), the bulk-color deposits) — NOT imported λ QCD. This is the “differences are data” step.
3. = observed (the scale-consistent $\{1, 20, 1028\}$). Elie’s lead (flagged reverse-read): $m_b/m_d \approx 1028$ sits on $N_c \cdot g^3 = 3 \cdot 7^3 = 1029$ (0.1%) — cleaner than 30^2 ; forward question: does the geometry give $N_c \cdot g^3$?

FORWARD LEAD (gate-independent, NOVEL structural) — CP from $N_c=3$ · LYRA

More than KM (which only *counts* $3 \rightarrow CP$): the geometric mechanism — 3 points at complex cube-root corners make a handed triangle (2 points = a real line, no handedness). The three = $\mathbb{Z}_3$ = colors = generations = corners; Casey: three sequential 2D-complex substrate commits composing the object. CP exists because $N_c=3$; matter/antimatter = the triangle orientation (same handedness as the CMB Axis of Evil). Bank IF the $\mathbb{Z}_3$ triality is shown FORCED (the generations genuinely carry the color- $\mathbb{Z}_3$ / sit at the cube-root corners) — not assumed. The value ($\sim 68^\circ$) is gated on the positions; the EXISTENCE + mechanism + handedness are gate-independent. Push it.

DEEP OPEN — α root-lock #2 · GRACE (if not on positions)

Does the S^2 coupling curvature reproduce the bulk Wyler volume ($\alpha^{-1} \sim \text{Vol}(D_{IV}^5)^{1/4}$, the $1920/\pi^5$ kernel) through the holomorphic extension? Confirms the trunk. Deep, genuinely open, her lane — don't fabricate.

Per-CI

- LYRA: force the positions (with Grace) + CP-from- $N_c=3$ (force the $\mathbb{Z}_3$ triality — the novel bank). The one gate + the jewel.
- GRACE: the positions + BST-native spectrum vs CONSTITUENT (her tell) + the native→observed dressing from confinement; OR α root-lock #2. Positions higher-leverage.
- ELIE: the native-scheme conversions (the dressing as DATA); the $N_c \cdot g^3$ b/d lead (forward-derive or reverse-read?); diagonalize when positions land.
- KEEPER: adjudicate the positions, the CP- $\mathbb{Z}_3$ candidate, the native-vs-constituent spectrum, the differences-as-data framing; track the count with Cal; over-sell #7.
- CAL: finalize the reconciled count (5 firm + α).

Disciplines armed

BST-native, not λ QCD — the difference is DATA to DERIVE (from confinement/Hamming), not error, not imported. $d(\nu)$ -native = a LEAD (Grace's tell), not a claim (b/d still open). CP-from- $N_c=3$ = novel candidate — show the $\mathbb{Z}_3$ triality forced (beautiful → hardest look). $N_c \cdot g^3$ = reverse-read lead (zero weight until forward). Force the positions (shown, not assumed). Compare to the SCALE-CONSISTENT / constituent target, never the scheme-fuzzy {1,20,900}. Over-sell #7 watch. Count as Cal reconciles.

What could bank (forward)

- The positions forced → native masses + mixing (close if native≈constituent; the MS-bar difference = data).
- CP-from- $N_c=3$ → novel structural bank (IF $\mathbb{Z}_3$ triality forced).
- α root-lock #2 → confirms the trunk.
- The dressing (native→observed) derived from confinement → the “differences are data” program lands.

Reserved for Casey

Mass-frame ratification · the paper (the “describe the BST SM, explain the differences as data” structure is a strong paper frame — gated on the reconciled count).

Long-pull target

Force the positions → the native spectrum (vs constituent) + mixing. Derive the native→observed dressing from BST's own confinement (differences = data). Land CP-from- $N_c=3$ (force the $\mathbb{Z}_3$). α root-lock #2 confirms the trunk. The masses are OPEN, worked in BST's own scheme, and the differences from observed are the geometry's data — exactly Casey's frame.

— Keeper, 2026-07-08. CASEY'S METHODOLOGY: describe the BST-native SM; differences from observed = DATA (the λ QCD dressing) to derive from BST's own confinement, not errors. The mass floor was a SCHEME error ($895 \times \text{MS-bar}$ vs $14 \times \text{constituent}$ = factor-60 λ QCD); $d(\nu)$ may be the native spectrum ($s/d=1.67 \approx \text{constituent}$). PRIMARY: force the positions → native masses (vs constituent) + mixing. NOVEL LEAD: CP-from- $N_c=3$ (geometric mechanism for KM; $\mathbb{Z}_3=\text{colors}=\text{generations}=\text{corners}$; handedness=matter/antimatter; force the triality). Compare to scale-consistent {1,20,1028} + $N_c \cdot g^3$ lead, never scheme-fuzzy {1,20,900}. α root-lock #2. Over-sell #7. Masses open in BST's own scheme.